

Exercise Sheet Week 8

Key:

- **Questions marked with a 🍷** are for discussion in your tutorials. Please complete these questions before your tutorial.
- **Questions marked with a 🦋** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🍷 **Exercise 1.** Given the function $f : [-3, 3] \rightarrow \mathbb{R}$, $f(x) = |(x^2 + 1)(x^2 - 4)|$.

- Set up a piecewise definition of f , not involving any absolute values.
- Find the local maxima and minima of f on its domain.
- Find the absolute maximum and the absolute minimum of f on its domain.

🍷 **Exercise 2.**

- Apply the Mean Value Theorem to $\arctan x$ to show that

$$\frac{b-a}{1+b^2} < \arctan(b) - \arctan(a) < \frac{b-a}{1+a^2}$$

where $0 < a < b$.

- Use the result of part (i) to obtain the value of $\arctan\left(\frac{21}{20}\right)$ to 2 decimal places of accuracy.

🍷 **Exercise 3.** (i) Sketch the curve $y = x^2 + x - 12$ and by completing the square, find the minimum value of y .

- Explain how you would use your sketch of $y = x^2 + x - 12$ to sketch the curve $y = \frac{1}{x^2 + x - 12}$ indicating the behaviour of y for large values of $|x|$, and the values of x for which y is not defined.

🎵 **Exercise 4.** (i) Prove that the equation $\frac{1}{2}x^4 - 3x^2 + 4x + 4 = 0$ has at least **one** solution in $[-3, 0]$.

- Prove that the equation $\frac{1}{2}x^4 - 3x^2 + 4x + 4 = 0$ has at least **two** solutions in $[-3, 0]$.

🎵 **Exercise 5.** Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

- $y = (\tanh(x))^x$
- $x^3 + \sin(xy) = xy^2$
- $x = 3 \cosh(2t)$, $y = 5 \sinh(2t)$.

♪ **Exercise 6.**

- (i) If $y = \left(x + \sqrt{1+x^2}\right)^{\frac{1}{2}}$, show that

$$(1+x^2) \left(\frac{dy}{dx}\right)^2 = \frac{1}{4}y^2.$$

- (ii) Hence show that

$$(1+x^2) \left(\frac{d^2y}{dx^2}\right) + x \frac{dy}{dx} - \frac{1}{4}y = 0.$$

♪ **Exercise 7.** (i) Find the 9th derivative of $\cos(x)$.

- (ii) Determine an expression for the $(4n+3)$ rd derivative of $\cos(x)$ and prove, by induction, that your expression is correct for all $n \geq 0$.

♪ **Exercise 8.** Use Leibniz rule to find

- (i) $\frac{d^3y}{dx^3}$ when $y = (x-2)^{1/2}(3x+4)^{1/3}$
- (ii) $\frac{d^4y}{dx^4}$ when $y = \frac{\ln(3x)}{x^2}$.

♪ **Exercise 9.** State L'Hôpital's rule and use it to evaluate the following limits.

- (i) $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x}$
- (ii) $\lim_{x \rightarrow 0^+} x^{\sin(x)}$
- (iii) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - x^2 - 2}{\sin^2(x) - x^2}$

♪ **Exercise 10.** Sketch the following curves indicating intersections with the axes and any stationary points and points of inflexion, if they exist:

- (i) $y = \frac{\ln x}{x}$
- (ii) $y = x^3 e^x$
- (iii) $y^2 = \frac{x^2 - 1}{x^2 + 1}$
- (iv) $y^2 = \frac{x^2 + 1}{x^2 - 1}$

♣ **Exercise 11.** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued function, and let $a < b$ and suppose $f'(a) < 0 < f'(b)$. In the following, you may assume f is continuous and differentiable, but you may not assume that f' is continuous.

- (i) If $f(a) < f(b)$ show there exists a point $d \in (a, b)$ with $f(d) = f(a)$.
- (ii) If $f(a) > f(b)$ show there exists a point $d \in (a, b)$ with $f(d) = f(b)$.
- (iii) Conclude there exists $c \in (a, b)$ for which $f'(c) = 0$. (Hint: Rolle's Theorem.)

♣ **Exercise 12.** State Leibniz rule for repeated differentiation of a product of two functions, then use proof by induction to prove Leibniz rule.

♣ **Exercise 13.** A tin can, consisting of a cylinder with circular disks top and bottom made of the same material, is to be fashioned using a fixed total area of sheet metal. Find the ratio of the tin's diameter, x , to its height, y , when its volume is a maximum and when its volume is a minimum. (Hint: let the total area of sheet metal to be used be 1. Find the volume V as a function of the diameter x .)